The empirical recurrence for OEIS A282835: recovering a conjecture that is not written down

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Abstract

OEIS A282835 records "Empirical recurrence of order 63", with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 116$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 63, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 63 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry's recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A282835 is "Number of nX5 0..1 arrays with no 1 equal to more than one of its king-move neighbors, with the exception of exactly two elements.". It has offset 1 and begins

$$2, 32, 1019, 17965, 215304, 2839946, 33527252, 380934118, \ \ldots$$

The entry states:

> Empirical recurrence of order 63 (see link above)

As of the "Last modified" line on the live entry (Feb 22 2017 11:20:23, revision 4) this is still recorded as empirical. The recurrence's coefficients are not in the entry text.

2 The count is a walk count

The entry's condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u, and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 810$ vertices, $a(n) = \iota^\top M^{j(n)}\tau$ for a linear reindexing j fixed by the entry's shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 116$ of the 810, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \ldots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' =$ 116.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 232$ exact terms of the model returns order 63 — the order the entry states — with the integer coefficients

$$a(n) \;=\; \sum_{i=1}^{63} c_i \, a(n-i),$$

c_1	12	c_{17}	4743610725	c_{33}	24209408258894	c_{49}	7355021193
c_2	45	c_{18}	−23266803591	c_{34}	−20460514525098	c_{50}	−5926904112
c_3	−299	c_{19}	60860922027	c_{35}	16903653301686	c_{51}	4688552726
c_4	−4752	c_{20}	−120023476455	c_{36}	−14084103149345	c_{52}	−385555068
c_5	−4020	c_{21}	330461495878	c_{37}	9780414656919	c_{53}	389857275
c_6	89248	c_{22}	−711771771822	c_{38}	−7121386335792	c_{54}	−388413090
c_7	630336	c_{23}	1395424375542	c_{39}	5320582047087	c_{55}	37081863
c_8	1011414	c_{24}	−2897254285716	c_{40}	−2963550391740	c_{56}	−2557494
c_9	−1490015	c_{25}	5100348298683	c_{41}	2039928652479	c_{57}	26596701
c_{10}	−17887047	c_{26}	−8180744847252	c_{42}	−1393006988912	c_{58}	−1517697
c_{11}	−22051512	c_{27}	12651822101892	c_{43}	581459809695	c_{59}	−313335
c_{12}	−11707516	c_{28}	−16935676229850	c_{44}	−424154101431	c_{60}	−857439
c_{13}	274815357	c_{29}	20800461868815	c_{45}	264128189674	c_{61}	161190
c_{14}	69895299	c_{30}	−24563871181942	c_{46}	−76314332073	c_{62}	56700
c_{15}	1003886138	c_{31}	25550324576511	c_{47}	63358668771	c_{63}	27000
c_{16}	−3928716048	c_{32}	−25066668285387	c_{48}	−38234716660		

Theorem 3. *The recurrence above holds for every n, and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 63, possibly only from some index on. Let q be that recurrence's characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $63+4$ terms removed returns order 63 as well, so $\deg q_{\min} = 63 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 17 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry's English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $63+4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 63 that this sequence satisfies — and that family turns out to have exactly one member.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A282835.

[2] J. L. Massey, Shift-register synthesis and BCH decoding, *IEEE Trans. Inform. Theory* **15** (1969), 122–127.

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011. (Transfer-matrix method, Section 4.7.)

[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.